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$$I = \int_{-\frac{\pi}{2}}^0 (5 \cos x - 3 \sin x) dx$$

Eerst vinden we de onbepaalde integraal

$$\begin{aligned} & \int (5 \cos x - 3 \sin x) dx \\ &= 5 \sin x + 3 \cos x + C \end{aligned}$$

Nu vullen we de grenzen in

$$\begin{aligned} I &= (5 \sin 0 + 3 \cos 0) - \left(5 \sin \left(-\frac{\pi}{2} \right) + 3 \cos \left(-\frac{\pi}{2} \right) \right) \\ &= 3 + 5 = 8 \end{aligned}$$

References